Placard main board

Hierarchy: each box is one sheet; the sheets connect through global labels

Power and clock

J13 USB-C, F1, U40 3.3 V

X1 25.175 MHz clock

STALL / RUN jumpers

File: power.kicad_sch

Control

U1 State, U2 Branch

U17 instruction register PLD

File: control.kicad_sch

Program counter and address mux

U3, U4 program counter

U7–U10 address mux

File: pc_mux.kicad_sch

Memory and I/O decode

U31, U32 RAM, J8 cartridge

U11 memory control

U12 I/O decode, U25 map

File: memory.kicad_sch

Video timing

U13 H, U14 V counters

U15 Sync, U35 audio slots

U29 line read

File: video_timing.kicad_sch

ALU and registers

U5, U6 ALU

U19 AC, U20 X, U21 Y

File: alu_regs.kicad_sch

Data bus

U18 D register

U26, U27 source drivers

R5–R12, RN1

File: data_bus.kicad_sch

Expansion slots

J9–J12 card-edge slots

File: expansion.kicad_sch

Video and audio out

U22 pixel buffer, U23 line code

U16 color decoder, VGA

U36 audio DAC, J6

File: video_out.kicad_sch

Shared port

U37 latch, U38 port PLD

U39 rows, J1–J5

File: shared_port.kicad_sch

34 chips + U40 regulator (docs/CHIPS.md numbers; U24, U28, U30, U33, U34 are not used).
Decoupling: every chip's 100 nF Cn sits beside Un on its own sheet.
Power: +5V, GND, +3V3 (SD card), GNDA (audio, joined to GND on sheet 2).
The ROM cartridge is its own project: kicad/cartridge/.

Chip numbers U1–U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

Placard computer

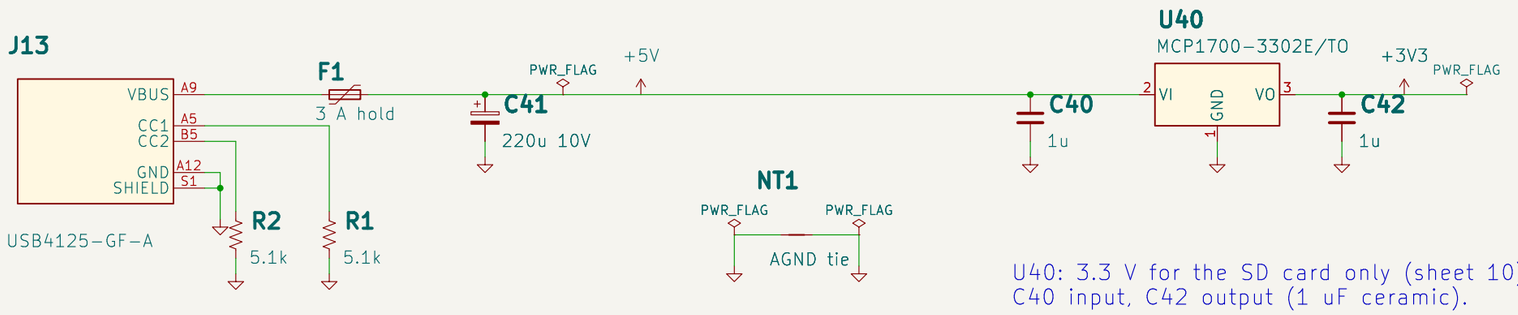
Sheet: /
File: placard.kicad_sch

Title: Placard: Main board

Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/11

Power and clock

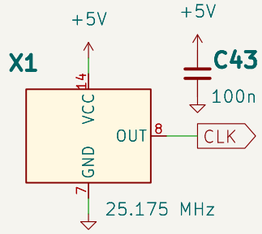
USB-C 5 V in, 3.3 V for the SD card, 25.175 MHz oscillator (CLK), cartridge STALL and RUN

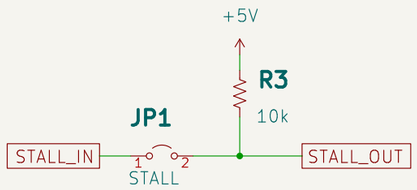


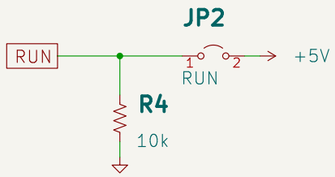
5.1k on CC1 and CC2: a USB-C source gives 5 V. 19 PLDs draw up to about 2.5 A (130 mA each at -7, worst case): use a 3 A supply; the polyfuse trips on a short.

AGND (audio) meets GND here, at the power input

U40: 3.3 V for the SD card only (sheet 10). C40 input, C42 output (1 uF ceramic).







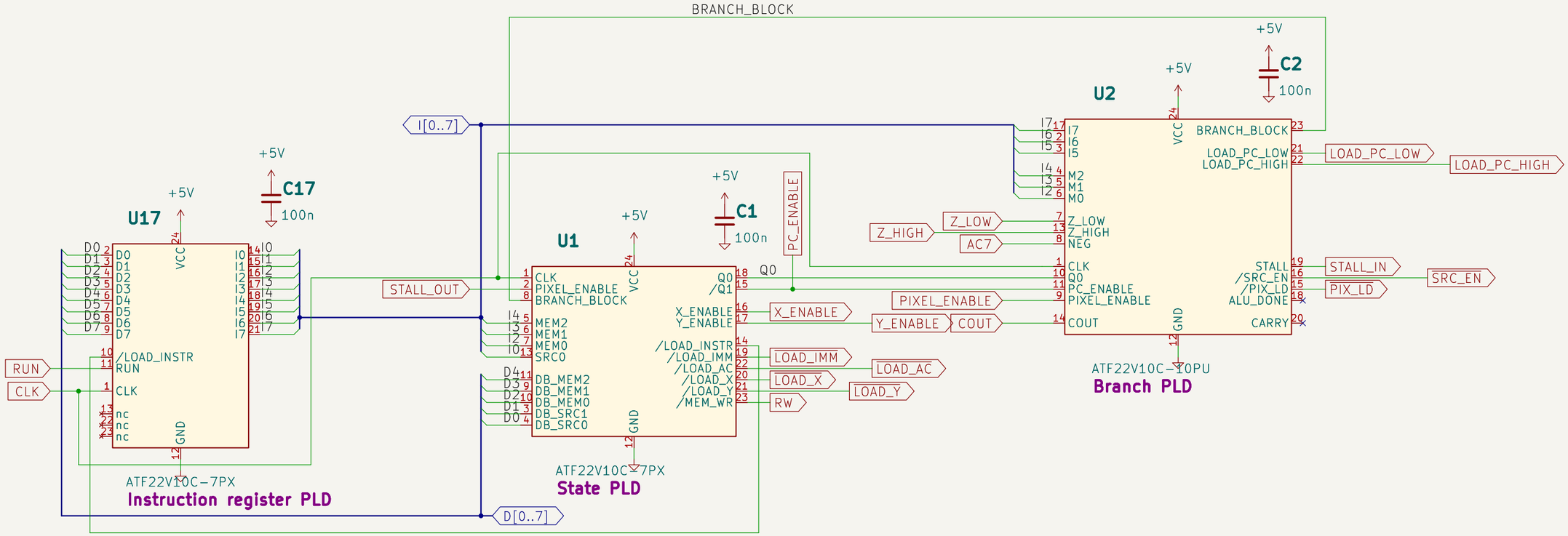
STALL_OUT (10k up) and RUN (10k down) come from the cartridge: STALL_IN (the Branch PLD's STALL) goes out to it. Without one, fit JP1 and JP2 for bring-up (BRINGUP.md): take them out with a cartridge.

Chip numbers U1-U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Power and clock/ File: power.kicad_sch		
Title: Placard: Power and clock		
Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/11

Control

State and Branch PLDs, instruction register



U17 loads the opcode in FETCH (/LOAD_INSTR) and holds 02h while RUN is low.
U2's STALL (STALL_IN) goes to the cartridge; STALL_OUT comes back to U1 and the PC.
U1 State: FETCH / LOAD / EXEC, register loads (active low, F377 /E), RW (1 = read).
U2 Branch: NEG is AC7. ALU_DONE (18) and CARRY (20) are internal registers: no connection.
Bring-up straps S1-S3 go in U1's socket (BRINGUP.md), not on the board.

Chip numbers U1-U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

Placard computer

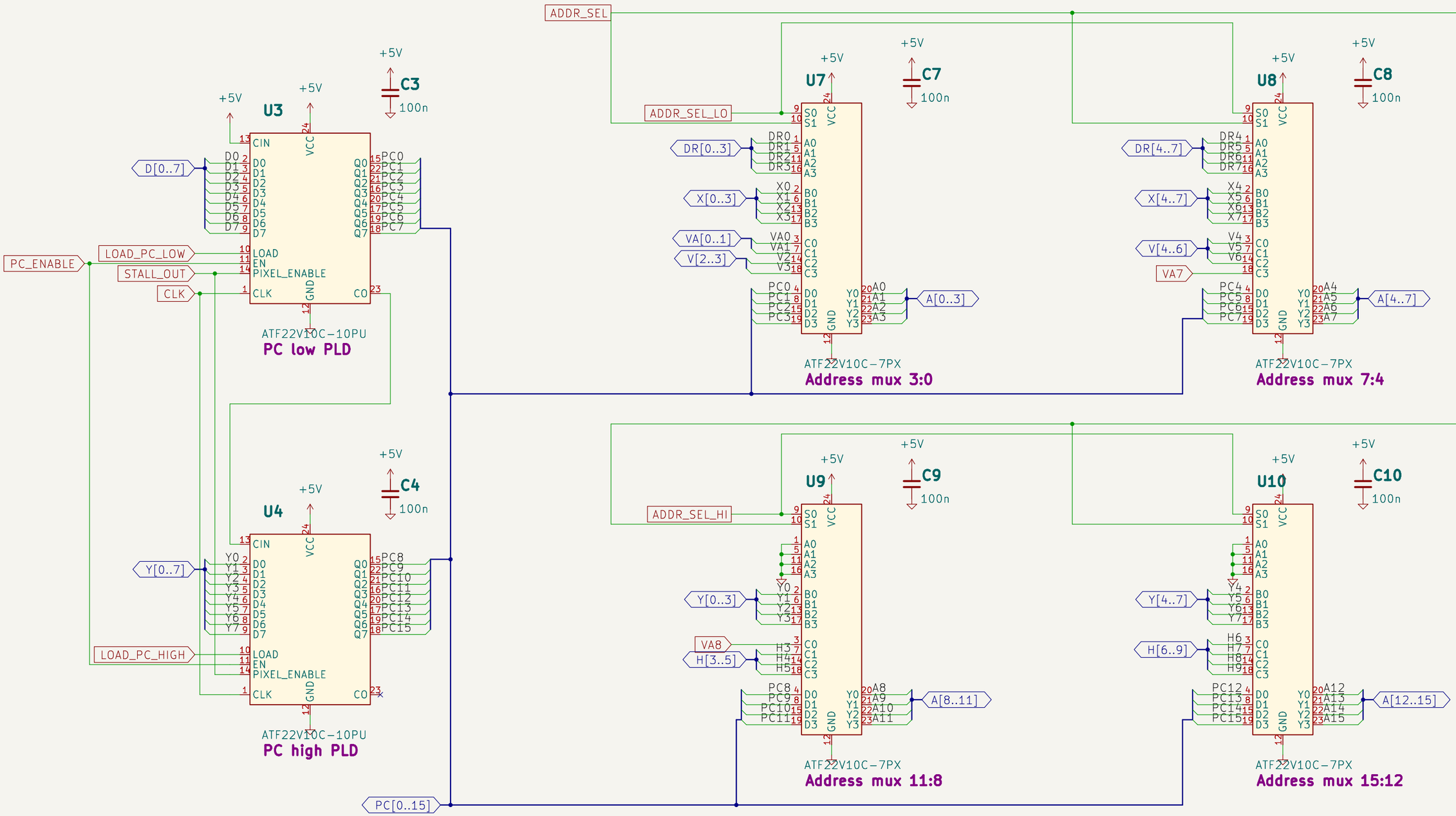
Sheet: /Control/
File: control.kicad_sch

Title: Placard: Control

Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/11

Program counter and address mux

Mux inputs: A = D / 0, B = X / Y, C = video address, D = PC



Video address: VA0, VA1, VA7, VA8 from the audio slot PLD (line repeat), A6:A2 = V6:V2, A15:A9 = H9:H3.
U3 CIN tied high (always counts); U3 CO carries into U4. STALL_OUT holds the PC (video reads, ALU wait, cartridge load).

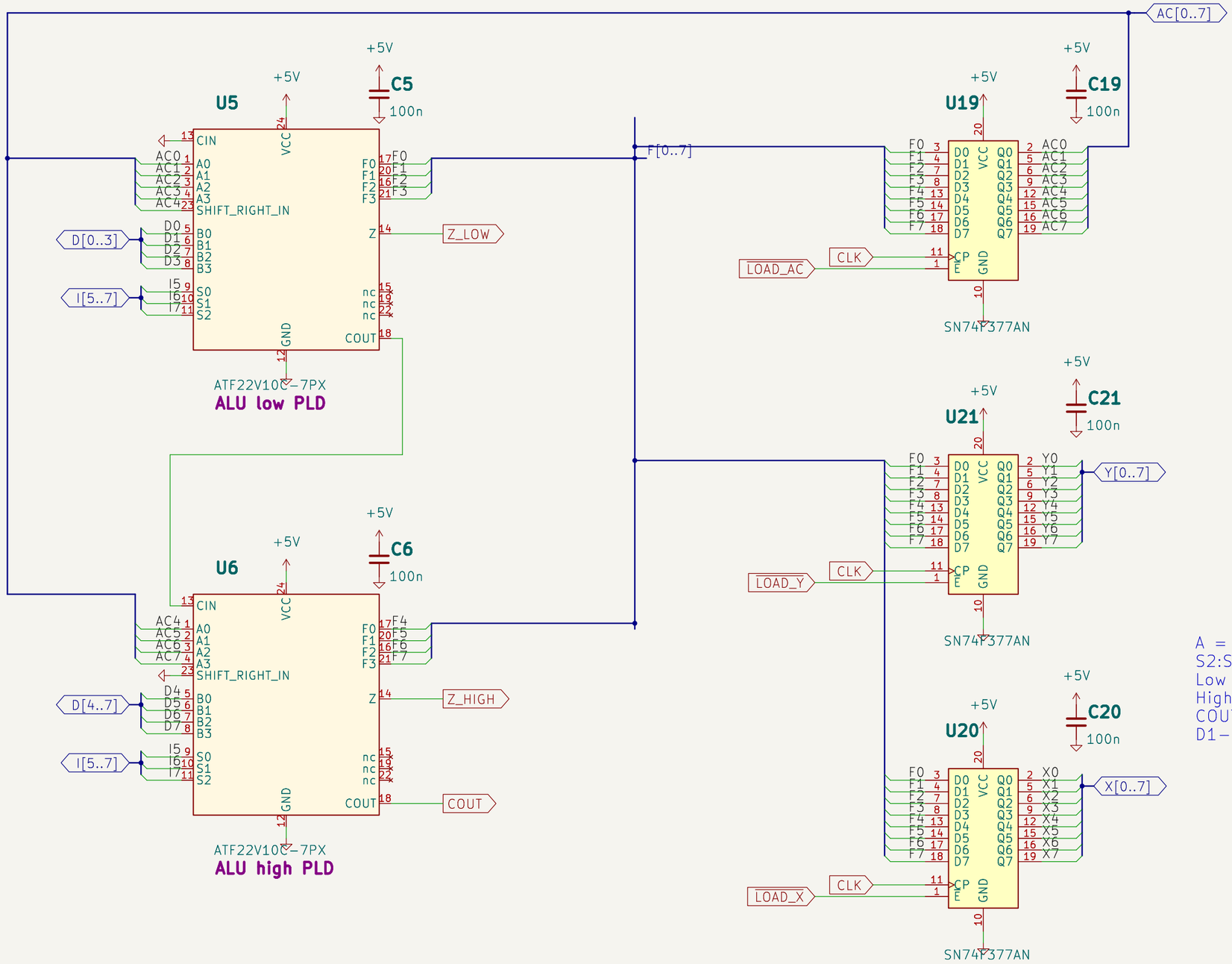
Chip numbers U1–U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer
Sheet: /Program counter and address mux/
File: pc_mux.kicad_sch

Title: Placard: Program counter and address mux

Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/11

ALU and registers

Two 4-bit ALU PLDs; F7:F0 loads AC, X or Y



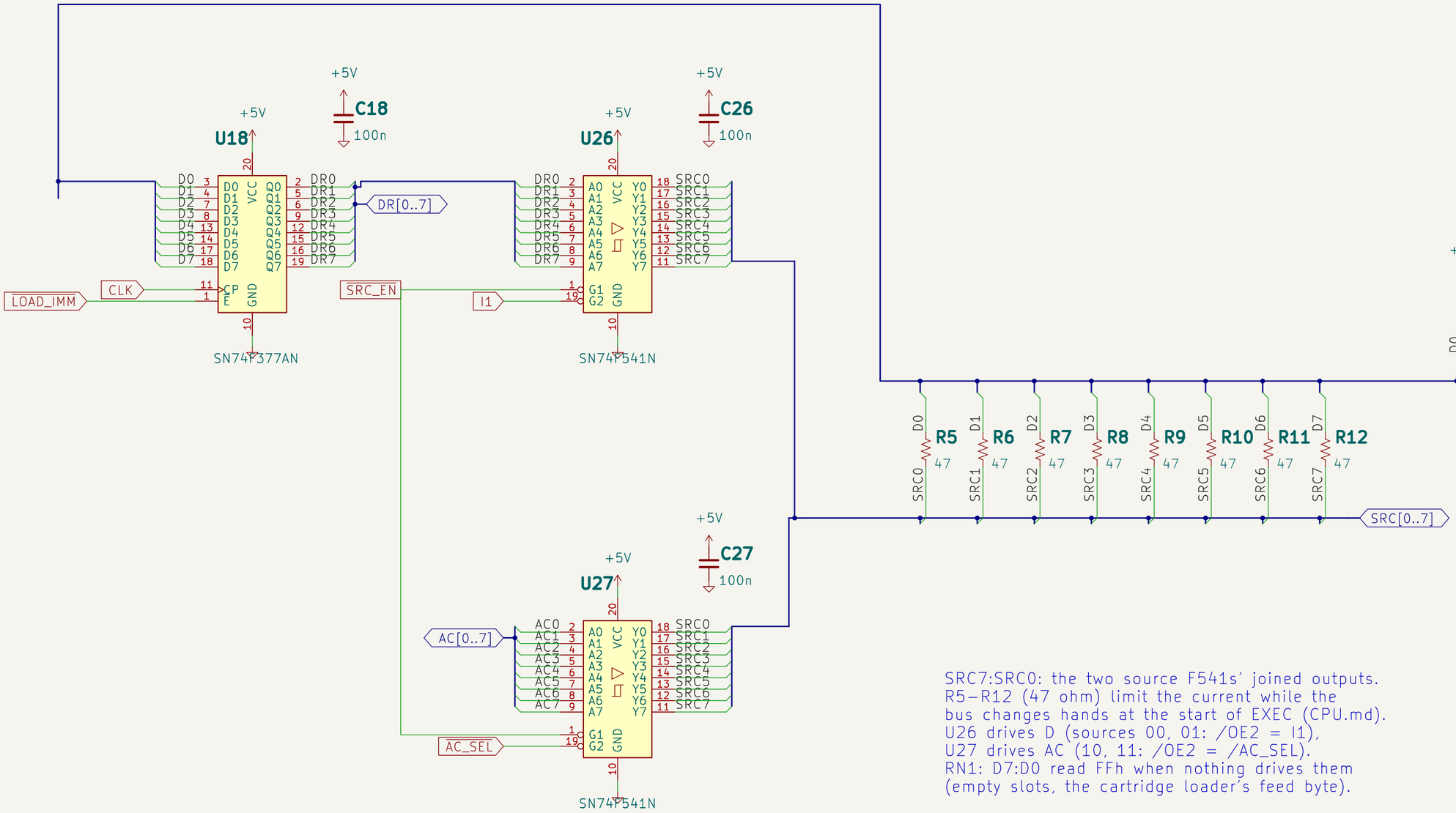
A = AC, B = the data bus (the source in EXEC).
S2:S0 = I7:I5.
Low nibble: CIN = 0, shift-right in = AC4.
High nibble: CIN = low COUT, shift-right in = 0.
COUT (high) is the carry for the Branch PLD.
D1–D3 (15, 19, 22) are internal carry nodes.

Chip numbers U1–U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /ALU and registers/		
File: alu_regs.kicad_sch		
Title: Placard: ALU and registers		
Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/11

Data bus

D register, source drivers (D or AC) onto the one data bus through series resistors, read pull-ups



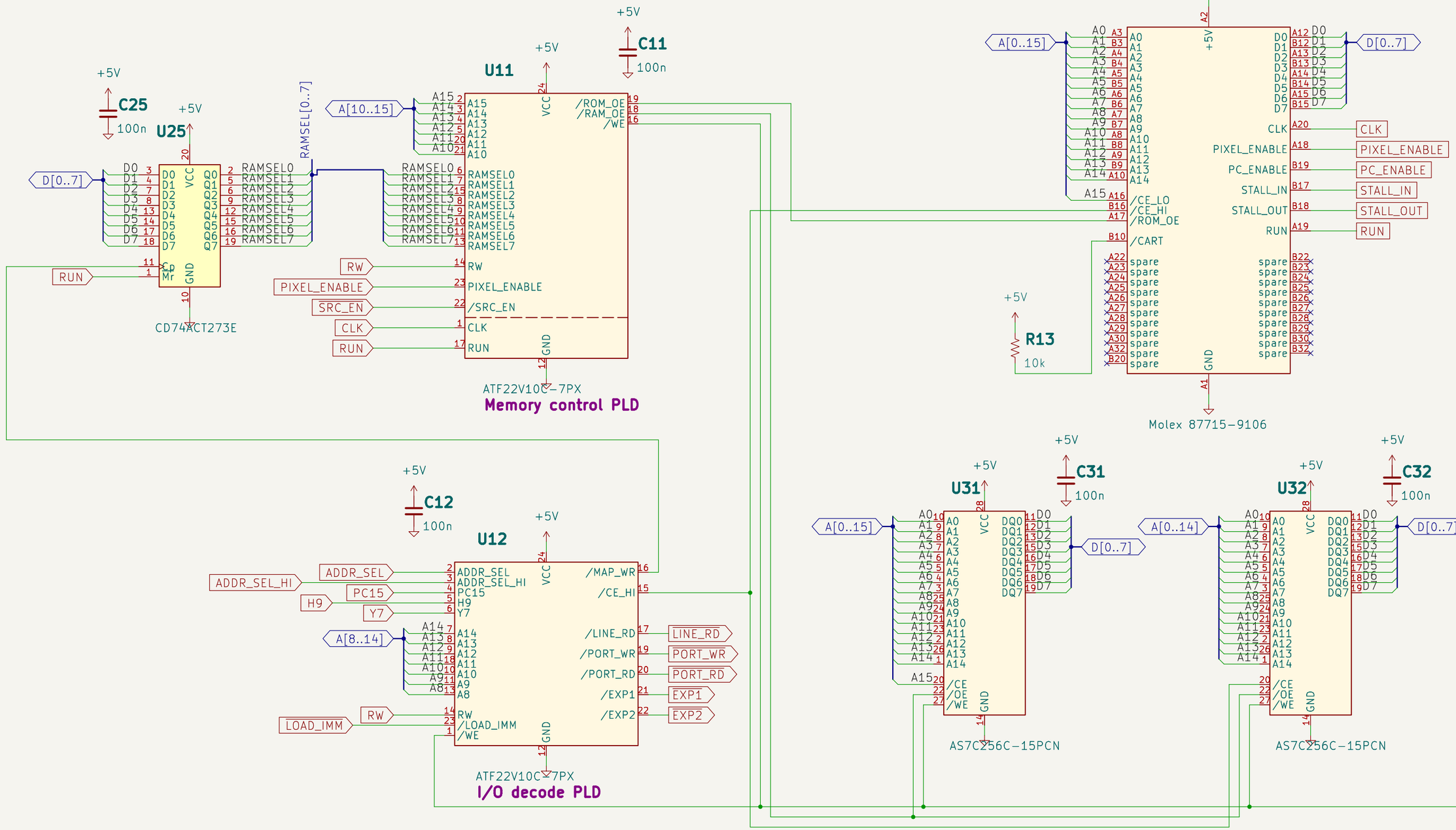
SRC7:SRC0: the two source F541s' joined outputs.
R5–R12 (47 ohm) limit the current while the
bus changes hands at the start of EXEC (CPU.md).
U26 drives D (sources 00, 01: /OE2 = I1),
U27 drives AC (10, 11: /OE2 = /AC_SEL).
RN1: D7:D0 read FFh when nothing drives them
(empty slots, the cartridge loader's feed byte).

Chip numbers U1–U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Data bus/		
File: data_bus.kicad_sch		
Title: Placard: Data bus		
Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/11

Memory and I/O decode

RAM, ROM cartridge slot, memory control, I/O decode, map register



Lower 32K: /CE = A15 (RAM U31, cartridge /CE_LO); upper: /CE_HI from U12.
RUN low (cartridge loading): /ROM_OE held off, map register clear (all ROM).
/CART reads low with a cartridge in; nothing reads it yet. 21 spare contacts.

Chip numbers U1–U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

Placard computer

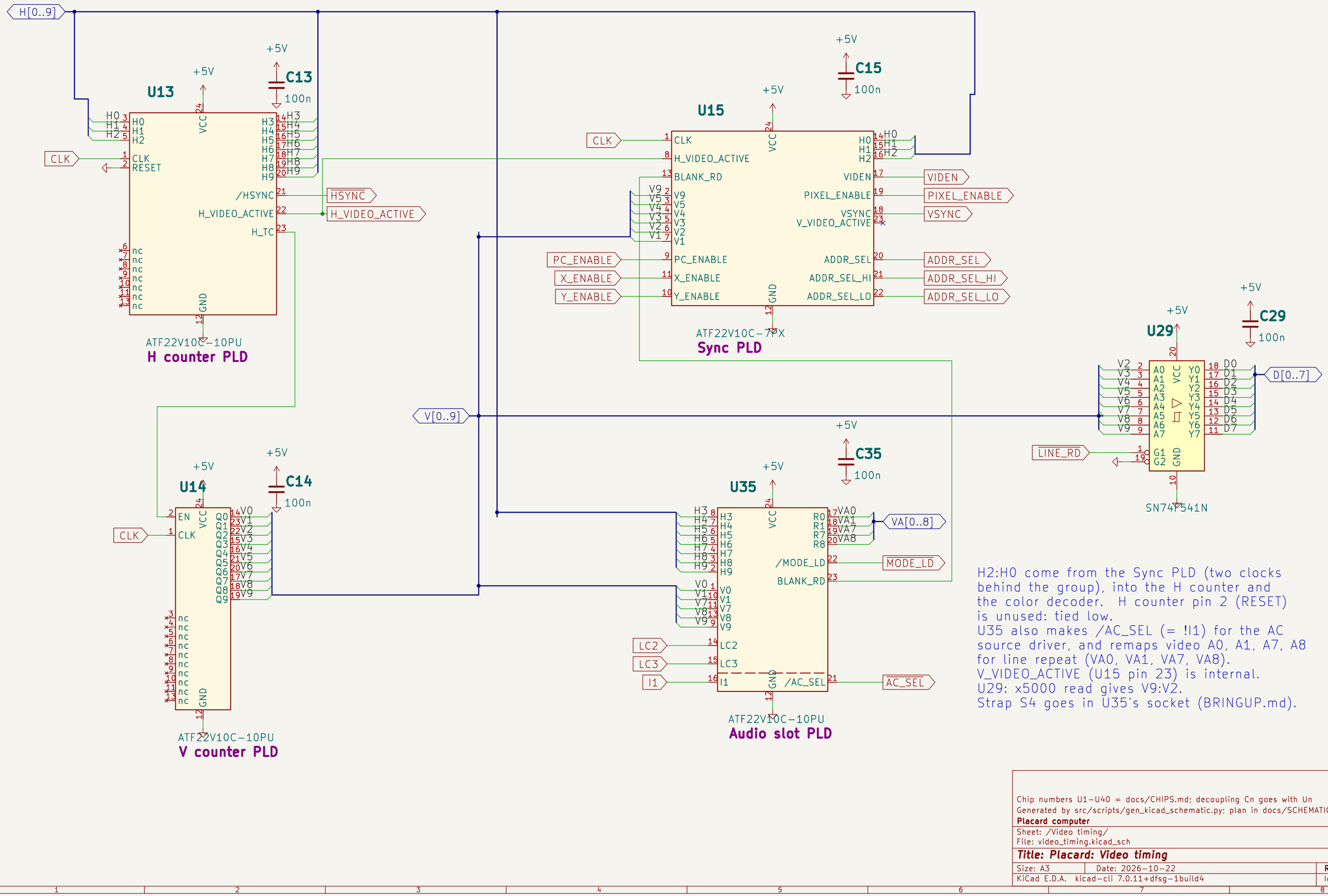
Sheet: /Memory and I/O decode/
File: memory.kicad_sch

Title: Placard: Memory and I/O decode

Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/11

Video timing

640 x 480 at 60 Hz: H and V counters, Sync, mode and audio slots, line read



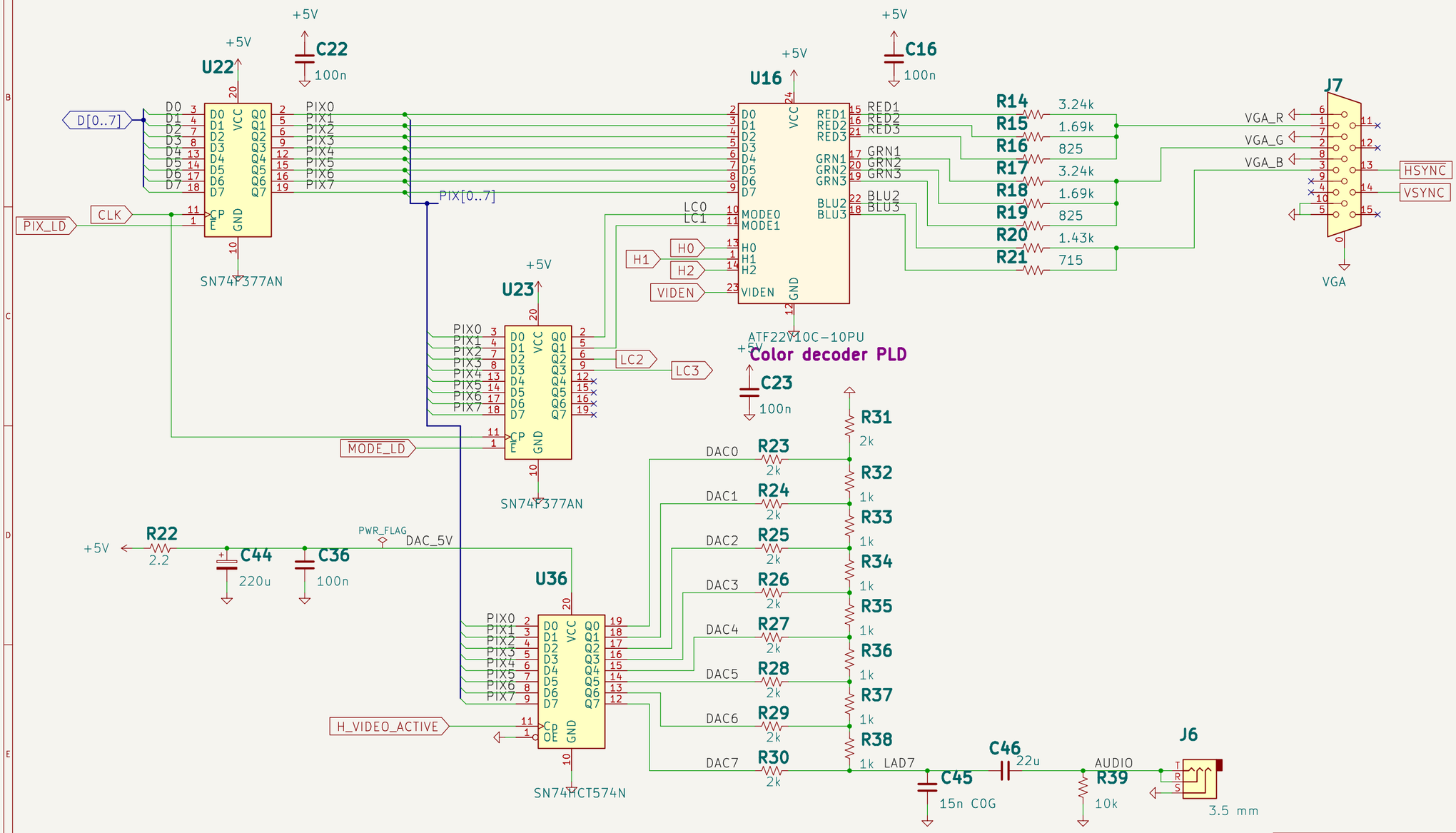
H2:H0 come from the Sync PLD (two clocks behind the group), into the H counter and the color decoder. H counter pin 2 (RESET) is unused: tied low.
U35 also makes /AC_SEL (= !I1) for the AC source driver, and remaps video A0, A1, A7, A8 for line repeat (VA0, VA1, VA7, VA8).
V_VIDEO_ACTIVE (U15 pin 23) is internal.
U29: x5000 read gives V9:V2.
Strap S4 goes in U35's socket (BRINGUP.md).

Chip numbers U1–U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Video timing/		
File: video_timing.kicad_sch		
Title: Placard: Video timing		
Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/11

Video and audio out

Pixel buffer, line code, color decoder and VGA DAC; audio DAC register, R–2R ladder, headphones



U23 Q3:Q2 = line repeat to U35 (sheet 8); Q1:Q0 = color mode. DAC_5V: U36's own filtered rail (2.2 ohm, 220u, C36).
Ladder: 2k 1 % per bit + 2k termination, 1k between nodes (LAD7 = out). AGND (GNDA) joins GND at the power input.

Chip numbers U1–U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

Placard computer

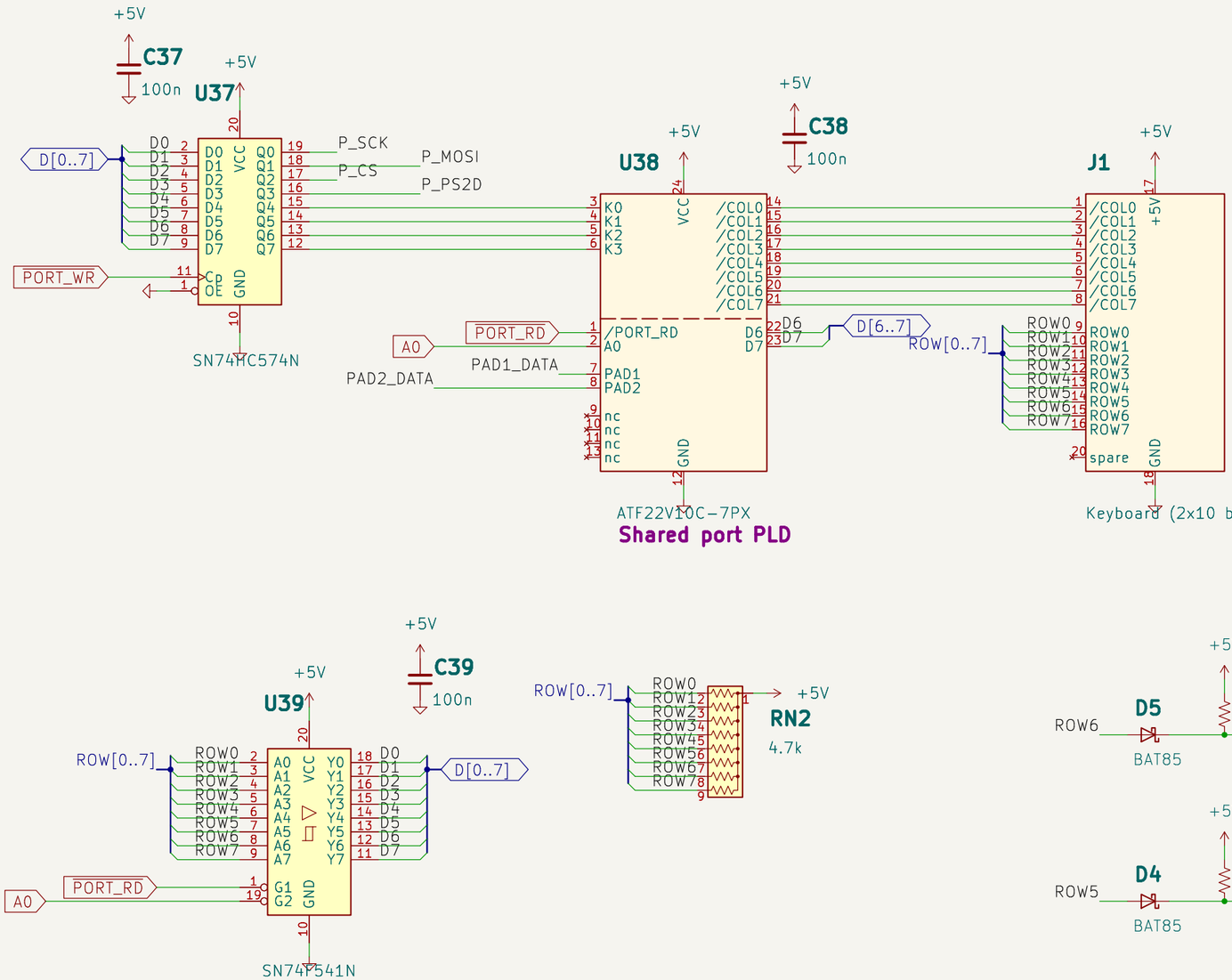
Sheet: /Video and audio out/
File: video_out.kicad_sch

Title: Placard: Video and audio out

Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/11

Shared port x5200

Keyboard, two SNES pads, SD card (SPI), PS/2 mouse



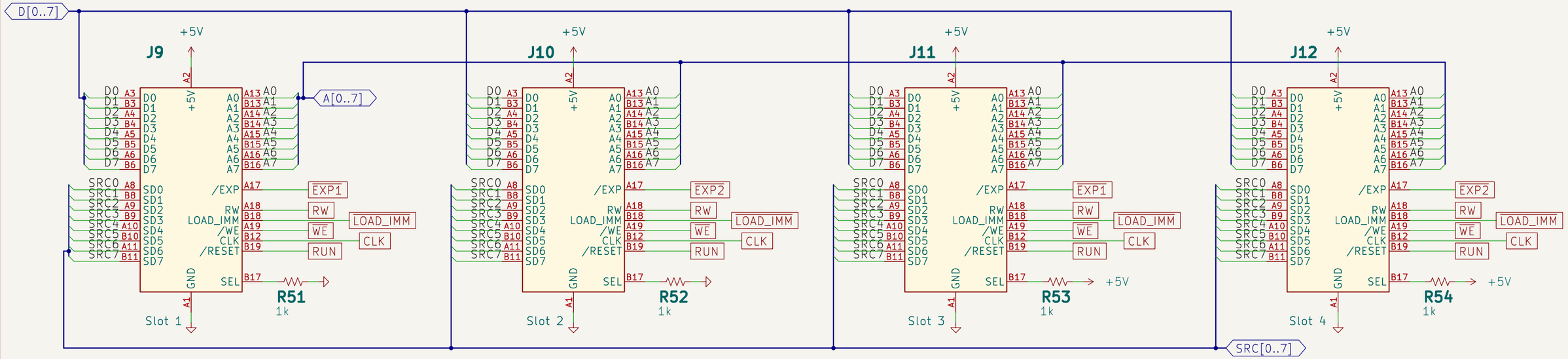
Latch Q0 = SCK and pad CLOCK, Q1 = MOSI and pad LATCH, Q2 = SD /CS (low also holds PS/2 CLK), Q3 = PS/2 DATA, Q7:Q4 = key column. SD lines come down to 3.2 V through 1.8k / 3.3k; MISO reaches row 7 through a Schottky.

Chip numbers U1–U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Shared port/		
File: shared_port.kicad_sch		
Title: Placard: Shared port		
Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 10/11

Expansion slots

Four 2 x 20 card-edge slots, x5100-x53FF, 128 bytes each (EXPANSION.md)



SEL: GND on slots 1 and 2, +5 V on slots 3 and 4 (through 1k): the A7 half the card answers.
/EXP1 goes to slots 1 and 3, /EXP2 to slots 2 and 4. /RESET = RUN (low while the cartridge loads).
SD7:SD0 = SRC7:SRC0, the source drivers' side of the series resistors (the write data).
LOAD_IMM carries the State PLD's /LOAD_IMM (low when the CPU latches the read bus into D).

Chip numbers U1-U40 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Expansion slots/
File: expansion.kicad_sch

Title: Placard: Expansion slots

Size: A3	Date: 2026-10-22	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 11/11